HW5

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Q1: 2.15 (c)

Consider the probability space $\big((0,1), \mathcal{B}((0,1)), m\big)$, where $m(\cdot)$ is the Lebesgue measure.

Let $F : \mathbb{R} \to \mathbb{R}$ be a cdf. For $0 < x < 1$, let

$$F_1^{-1}(x) = \inf\{y : y \in \mathbb{R},\ F(y) \geq x\}$$
$$F_2^{-1}(x) = \sup\{y : y \in \mathbb{R},\ F(y) \leq x\}.$$

Let Z_i be the random variable defined by

$$Z_i = F_i^{-1}(x), \quad 0 < x < 1, \quad i = 1, 2.$$

We utilize the following results for (i) and (ii):

- Definition 2.2.4 (cdf): $F : \mathbb{R} \to \mathbb{R}$ is a cdf if it is nondecreasing, right continuous (i.e., $F(x) = \lim_{y \downarrow x} F(y)$), and satisfies $\lim_{x \to -\infty} F(x) = 0$ and $\lim_{x \to \infty} F(x) = 1$, the properties of Proposition 2.2.2.
- Definition 2.1.2(b) (random variable): $X : \Omega \to \mathbb{R}$ is a random variable if $\{\omega : X(\omega) \leq a\} \in \mathcal{F}$ for each $a \in \mathbb{R}$.
- Definition 2.2.3 (cdf of a random variable): $F_X(t) \equiv P(X \leq t)$, $t \in \mathbb{R}$.
- Lebesgue measure: m of an interval is its length, so $m\big((0,c) \cap (0,1)\big) = m\big((0,c] \cap (0,1)\big) = c$ for $0 \leq c \leq 1$; and $A \subset B$ implies $m(A) \leq m(B)$.

(i)

Find the cdf of Z_i, $i = 1, 2$.

(Hint: Verify using the right continuity of F that for any $0 < x < 1$, $t \in \mathbb{R}$, $F(t) \geq x \iff F_1^{-1}(x) \leq t$.)

Answer

Z_1 and Z_2 both have cdf F.

For $x \in (0,1)$, let $A_x \equiv \{y \in \mathbb{R} : F(y) \geq x\}$ and $B_x \equiv \{y \in \mathbb{R} : F(y) \leq x\}$, so that $F_1^{-1}(x) = \inf A_x$ and $F_2^{-1}(x) = \sup B_x$. Both are finite. Since F is nondecreasing with $F(y) \to 0 < x$ as $y \to -\infty$ and $F(y) \to 1 > x$ as $y \to \infty$ (Definition 2.2.4), A_x is nonempty and bounded below (by any y_0 with $F(y_0) < x$), and B_x is nonempty and bounded above (by any y_1 with $F(y_1) > x$).

Proof.

Claim (1): For any $x \in (0,1)$ and $t \in \mathbb{R}$,

$$F(t) \geq x \iff F_1^{-1}(x) \leq t. \tag{1}$$

Let $x \in (0,1)$ and $t \in \mathbb{R}$ be given, and let $q \equiv F_1^{-1}(x)$.

($\Rightarrow$) Suppose $F(t) \geq x$. Then $t \in A_x$, so $q = \inf A_x \leq t$.

($\Leftarrow$) Suppose $q \leq t$. We first show $F(q) \geq x$. For each $n \geq 1$, $q + n^{-1}$ is not a lower bound of A_x, so there is a $y_n \in A_x$ with $q \leq y_n < q + n^{-1}$. Then $y_n \to q$ with $y_n \geq q$, so $F(y_n) \to F(q)$ by the right continuity of F. Since $F(y_n) \geq x$ for all n, $F(q) \geq x$. Since F is nondecreasing and $q \leq t$, $F(t) \geq F(q) \geq x$.

cdf of Z_1: Let $t \in \mathbb{R}$ be given. By (1),

$$\{x \in (0,1) : Z_1(x) \leq t\} = \{x \in (0,1) : x \leq F(t)\} = (0, F(t)] \cap (0,1).$$

This is an interval, so it is in $\mathcal{B}((0,1))$, and Z_1 is a random variable (Definition 2.1.2(b)). Since $0 \leq F(t) \leq 1$, its Lebesgue measure is $F(t)$. Thus $P(Z_1 \leq t) = F(t)$.

cdf of Z_2: The equivalence (1) can fail for F_2^{-1}. For instance, if $F \equiv x$ on an interval $[a, b]$ and $F > x$ beyond b, then $F(a) \geq x$ but $F_2^{-1}(x) = b > a$. Instead, we claim that for any $x \in (0,1)$ and $t \in \mathbb{R}$,

$$F(t) > x \implies F_2^{-1}(x) \leq t \implies F(t) \geq x. \tag{2}$$

For the first implication, suppose $F(t) > x$, and let $y \in B_x$. Then $F(y) \leq x < F(t)$, so $y < t$ since F is nondecreasing. Thus t is an upper bound of B_x, and $F_2^{-1}(x) = \sup B_x \leq t$.

For the second, suppose, for contradiction, that $F_2^{-1}(x) \leq t$ but $F(t) < x$. By the right continuity of F, there is an $s > t$ with $F(s) < x$. Then $s \in B_x$, so $F_2^{-1}(x) \geq s > t$, a contradiction.

Now let $t \in \mathbb{R}$ be given. By (2),

$$\{x \in (0,1) : x < F(t)\} \subset \{x \in (0,1) : F_2^{-1}(x) \leq t\} \subset \{x \in (0,1) : x \leq F(t)\}.$$

The two outer sets are $(0, F(t)) \cap (0,1)$ and $(0, F(t)] \cap (0,1)$. They differ by at most the point $F(t)$, and each has Lebesgue measure $F(t)$. So the middle set is one of the two outer sets, hence is in $\mathcal{B}((0,1))$, and Z_2 is a random variable. By the monotonicity of m,

$$P(Z_2 \leq t) = m\big(\{x \in (0,1) : F_2^{-1}(x) \leq t\}\big) = F(t).$$

Since t was arbitrary, Z_1 and Z_2 both have cdf F (Definition 2.2.3). Said differently, starting from Lebesgue measure on $(0,1)$, Z_1 (or Z_2) produces a random variable with any prescribed cdf F. //

(ii)

Show also that $F_1^{-1}(\cdot)$ is left continuous and $F_2^{-1}(\cdot)$ is right continuous.

Answer

Proof. Let A_x and B_x be as in (i). Both functions are nondecreasing on $(0,1)$. For $0 < x \leq x' < 1$, $A_{x'} \subset A_x$ and $B_x \subset B_{x'}$, so

$$F_1^{-1}(x) = \inf A_x \leq \inf A_{x'} = F_1^{-1}(x') \quad \text{and} \quad F_2^{-1}(x) = \sup B_x \leq \sup B_{x'} = F_2^{-1}(x').$$

Let $x \in (0,1)$ and $\epsilon > 0$ be given.

Left continuity of F_1^{-1}: Let $q \equiv F_1^{-1}(x)$. Since $q - \epsilon < \inf A_x$, $q - \epsilon \notin A_x$, i.e., $F(q - \epsilon) < x$. Set $\delta \equiv x - F(q - \epsilon) > 0$, and let $x' \in (x - \delta, x)$. Then $0 \leq F(q - \epsilon) < x' < x < 1$, so $x' \in (0,1)$. Since $F(q - \epsilon) < x'$, (1) applied to x' and $t = q - \epsilon$ gives $F_1^{-1}(x') > q - \epsilon$. Also $F_1^{-1}(x') \leq q$, since F_1^{-1} is nondecreasing. Hence

$$0 \leq F_1^{-1}(x) - F_1^{-1}(x') < \epsilon \quad \text{for all } x' \in (x - \delta, x).$$

Right continuity of F_2^{-1}: Let $s \equiv F_2^{-1}(x)$. Since $s + \epsilon > \sup B_x$, $s + \epsilon \notin B_x$, i.e., $F(s + \epsilon) > x$. Set $\delta \equiv F(s+\epsilon) - x > 0$, and let $x' \in (x, x + \delta)$. Then $0 < x < x' < F(s+\epsilon) \leq 1$, so $x' \in (0,1)$. Since $F(s+\epsilon) > x'$,

the first implication in (2), applied to x' and $t = s + \epsilon$, gives $F_2^{-1}(x') \le s + \epsilon$. Also $F_2^{-1}(x') \ge s$, since F_2^{-1} is nondecreasing. Hence

$$0 \le F_2^{-1}(x') - F_2^{-1}(x) \le \epsilon \quad \text{for all } x' \in (x, x+\delta).$$

Since $x \in (0,1)$ and $\epsilon > 0$ were arbitrary, F_1^{-1} is left continuous and F_2^{-1} is right continuous on $(0,1)$. //

Q2: 2.20

Apply Corollary 2.3.5 to show that for any collection $\{a_{ij} : i, j \in \mathbb{N}\}$ of nonnegative numbers,

$$\sum_{i=1}^{\infty}\left(\sum_{j=1}^{\infty} a_{ij}\right) = \sum_{j=1}^{\infty}\left(\sum_{i=1}^{\infty} a_{ij}\right).$$

Answer

We utilize the following results:

- Counting measure: on $(\mathbb{N}, \mathcal{P}(\mathbb{N}))$, $\mu(A)$ is the number of elements of A; in particular, $\mu(\{k\}) = 1$.
- Definition 2.3.2 (integral of a simple nonnegative function): if $f = \sum_{i=1}^{k} c_i I_{A_i}$ with $c_i \in [0,\infty]$ and $A_i \in \mathcal{F}$, then $\int f\, d\mu = \sum_{i=1}^{k} c_i\, \mu(A_i)$, with the convention $0 \cdot \infty = 0$.
- Corollary 2.3.5: for nonnegative measurable $\{h_n\}_{n\ge1}$, $\int \left(\sum_{n=1}^{\infty} h_n\right) d\mu = \sum_{n=1}^{\infty} \int h_n\, d\mu$.
- Remark 2.3.2: on a countable $\Omega = \{\omega_i : i \ge 1\}$ with $\mathcal{F} = \mathcal{P}(\Omega)$, every $f : \Omega \to [0,\infty)$ is measurable, and $\int f\, d\mu = \sum_{i=1}^{\infty} f(\omega_i)\, \mu(\{\omega_i\})$.

Proof. Let $\Omega \equiv \mathbb{N}$, $\mathcal{F} \equiv \mathcal{P}(\mathbb{N})$, and let μ be the counting measure on $(\Omega, \mathcal{F})$. Since $\mathcal{F}$ contains every subset of $\mathbb{N}$, every function on $\mathbb{N}$ is $\mathcal{F}$-measurable.

Lemma. If $h : \mathbb{N} \to [0,\infty]$, then $\int h\, d\mu = \sum_{k=1}^{\infty} h(k)$.

This is Remark 2.3.2 for counting measure, allowing the value ∞; we derive it from Corollary 2.3.5. For $k \in \mathbb{N}$, let $g_k \equiv h(k) I_{\{k\}}$. Each g_k is nonnegative and simple, so by Definition 2.3.2,

$$\int g_k\, d\mu = h(k)\, \mu(\{k\}) = h(k).$$

Also $h = \sum_{k=1}^{\infty} g_k$ pointwise, since for each $j \in \mathbb{N}$, $g_j(j) = h(j)$ and $g_k(j) = 0$ for $k \ne j$ (using $0 \cdot \infty = 0$). By Corollary 2.3.5,

$$\int h\, d\mu = \sum_{k=1}^{\infty} \int g_k\, d\mu = \sum_{k=1}^{\infty} h(k).$$

Now for $i \in \mathbb{N}$, define $h_i : \mathbb{N} \to [0,\infty)$ by $h_i(j) \equiv a_{ij}$, and let $h \equiv \sum_{i=1}^{\infty} h_i$, so that $h(j) = \sum_{i=1}^{\infty} a_{ij} \in [0,\infty]$. By the lemma,

$$\int h_i\, d\mu = \sum_{j=1}^{\infty} a_{ij} \quad \text{and} \quad \int h\, d\mu = \sum_{j=1}^{\infty}\left(\sum_{i=1}^{\infty} a_{ij}\right).$$

Since each h_i is nonnegative and measurable, Corollary 2.3.5 gives

$$\sum_{j=1}^{\infty}\left(\sum_{i=1}^{\infty} a_{ij}\right) = \int \left(\sum_{i=1}^{\infty} h_i\right) d\mu = \sum_{i=1}^{\infty} \int h_i\, d\mu = \sum_{i=1}^{\infty}\left(\sum_{j=1}^{\infty} a_{ij}\right). \quad //$$

Q3: 2.25

If $f(x) = I_{Q_1}(x)$ where $Q_1 = \mathbb{Q} \cap [0,1]$, $\mathbb{Q}$ being the set of all rationals, then show that for any partition P, $U(P,f) = 1$ and $L(P,f) = 0$.

Answer

We utilize the following results:

- Definition 2.4.1 (upper and lower Riemann sums): for a partition $P = \{x_0, \ldots, x_n\}$ of $[a, b]$, $U(f, P) \equiv \sum_{i=0}^{n-1} M_i(x_{i+1} - x_i)$ and $L(f, P) \equiv \sum_{i=0}^{n-1} m_i(x_{i+1} - x_i)$, where M_i and m_i are the sup and inf of f on $[x_i, x_{i+1}]$.
- Definitions 2.4.2 and 2.4.3 (Riemann integrability): f is Riemann integrable iff $\overline{\int} f \equiv \inf_P U(f, P)$ equals $\underline{\int} f \equiv \sup_P L(f, P)$.
- Definition 2.3.2 (integral of a simple nonnegative function): if $f = \sum_{i=1}^{k} c_i I_{A_i}$ with $c_i \geq 0$ and $A_i \in \mathcal{F}$, then $\int f \, d\mu = \sum_{i=1}^{k} c_i \, \mu(A_i)$.

Proof. Let $P = \{x_0, x_1, \ldots, x_n\}$ be a finite partition of $[0, 1]$, i.e., $0 = x_0 < x_1 < \cdots < x_n = 1$. As in Definition 2.4.1, for $i = 0, 1, \ldots, n-1$, let

$$M_i = \sup\{f(x) : x_i \leq x \leq x_{i+1}\} \quad \text{and} \quad m_i = \inf\{f(x) : x_i \leq x \leq x_{i+1}\}.$$

Fix i. Since $x_i < x_{i+1}$, the density of $\mathbb{Q}$ and of $\mathbb{R} \setminus \mathbb{Q}$ in $\mathbb{R}$ gives a rational r and an irrational s in $[x_i, x_{i+1}] \subset [0, 1]$. Then $r \in Q_1$ and $s \notin Q_1$, so $f(r) = 1$ and $f(s) = 0$. Since $0 \leq f \leq 1$, it follows that $M_i = 1$ and $m_i = 0$.

By Definition 2.4.1,

$$U(P, f) = \sum_{i=0}^{n-1} M_i \, (x_{i+1} - x_i) = \sum_{i=0}^{n-1} (x_{i+1} - x_i) = x_n - x_0 = 1$$

and

$$L(P, f) = \sum_{i=0}^{n-1} m_i \, (x_{i+1} - x_i) = 0.$$

Since P was arbitrary, $U(P, f) = 1$ and $L(P, f) = 0$ for any partition P.

Said differently, $\overline{\int} f = 1 \neq 0 = \underline{\int} f$, so f is not Riemann integrable (Definition 2.4.3). Yet f is Lebesgue integrable, with $\int f \, dm = m(Q_1) = 0$ by Definition 2.3.2, since Q_1 is countable and each singleton has Lebesgue measure 0. //

Q4: Extra (A)

Let $f : \Omega \to \mathbb{R}$ be a $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$-measurable nonnegative function. Define, for $n \geq 1$ and $x \in \Omega$,

$$f_n(x) = \begin{cases} \dfrac{i-1}{2^n}, & \text{if } \dfrac{i-1}{2^n} \leq f(x) < \dfrac{i}{2^n} \text{ for } i = 1, \ldots, n2^n \\ n, & \text{if } f(x) \geq n. \end{cases}$$

Then prove the following:

We utilize the following results for (i)–(iii):

- Definition 2.1.3 (measurable map): $T : \Omega_1 \to \Omega_2$ is $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$-measurable if $T^{-1}(A) \in \mathcal{F}_1$ for all $A \in \mathcal{F}_2$.
- Corollary 2.1.4: the $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$-measurable functions from Ω to $\mathbb{R}$ are closed under pointwise addition and multiplication, and under scalar multiplication.
- Proposition A.2.1 (Axiom of Archimedes): for every $y \in \mathbb{R}$, there exists $n \in \mathbb{N}$ with $n > y$.

(i)

For each $n \geq 1$, $f_n(x)$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$-measurable.

Answer

Proof. Let $n \geq 1$ be given, and let

$$A_{n,i} \equiv f^{-1}\Big(\Big[\tfrac{i-1}{2^n}, \tfrac{i}{2^n}\Big)\Big), \quad i = 1, \ldots, n2^n, \quad \text{and} \quad B_n \equiv f^{-1}\big([n, \infty)\big).$$

The intervals $\big[\frac{i-1}{2^n}, \frac{i}{2^n}\big)$, $i = 1, \ldots, n2^n$, and $[n, \infty)$ are disjoint with union $[0, \infty)$. Since $f \geq 0$, each $x \in \Omega$ lies in exactly one of $A_{n,1}, \ldots, A_{n,n2^n}, B_n$, and so

$$f_n = \sum_{i=1}^{n2^n} \frac{i-1}{2^n} I_{A_{n,i}} + n\, I_{B_n}.$$

Each of these intervals is in $\mathcal{B}(\mathbb{R})$ and f is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R})\rangle$-measurable, so $A_{n,i} \in \mathcal{F}$ for all i and $B_n \in \mathcal{F}$ (Definition 2.1.3).

For any $A \in \mathcal{F}$, I_A is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R})\rangle$-measurable, since $I_A^{-1}(B) \in \{\emptyset, A, A^c, \Omega\} \subset \mathcal{F}$ for all $B \in \mathcal{B}(\mathbb{R})$. Thus f_n is a finite linear combination of $\langle \mathcal{F}, \mathcal{B}(\mathbb{R})\rangle$-measurable functions, and so is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R})\rangle$-measurable by Corollary 2.1.4. //

(ii)

For each $x \in \Omega$, $f_n(x)$ is increasing with n (i.e., $\{f_n(\cdot)\}$ is an increasing sequence of functions).

Answer

Proof. For $n \geq 1$, let $P_n \equiv \{j/2^n : j = 0, 1, \ldots, n2^n\}$, the grid of mesh 2^{-n} on $[0, n]$. We first claim that for all $n \geq 1$ and $x \in \Omega$,

$$f_n(x) = \max\big\{p \in P_n : p \leq \min\{f(x), n\}\big\}. \tag{3}$$

That is, $f_n(x)$ is the largest grid point at or below $\min\{f(x), n\}$. Since $f \geq 0$, there are two cases.

Case 1: $f(x) \geq n$. Then $\min\{f(x), n\} = n = \max P_n$, so the right side of (3) is $n = f_n(x)$.

Case 2: $\frac{i-1}{2^n} \leq f(x) < \frac{i}{2^n}$ for some $i \in \{1, \ldots, n2^n\}$. Then $f(x) < n$, so $\min\{f(x), n\} = f(x)$. Also $j/2^n \leq f(x)$ iff $j \leq i-1$, so the right side of (3) is $(i-1)/2^n = f_n(x)$.

In either case, (3) holds.

With (3) in hand, let $x \in \Omega$ and $n \geq 1$ be given. Each $j/2^n \in P_n$ equals $2j/2^{n+1}$ with $2j \leq n2^{n+1} \leq (n+1)2^{n+1}$, so $P_n \subset P_{n+1}$. Also $\min\{f(x), n\} \leq \min\{f(x), n+1\}$. So

$$\big\{p \in P_n : p \leq \min\{f(x), n\}\big\} \subset \big\{p \in P_{n+1} : p \leq \min\{f(x), n+1\}\big\},$$

and taking the maximum of each side, $f_n(x) \leq f_{n+1}(x)$ by (3).

Since x and n were arbitrary, $f_n(x) \leq f_{n+1}(x)$ for all $x \in \Omega$ and $n \geq 1$. //

(iii)

Prove that $f_n(x) \to f(x)$ pointwise (i.e., for each fixed x), as $n \to \infty$.

Answer

Proof. Let $x \in \Omega$ be given. Since $f(x) \in [0, \infty)$, Proposition A.2.1 gives an $n_0 \in \mathbb{N}$ with $n_0 > f(x)$. For $n \geq n_0$, $0 \leq f(x) < n$, so $\frac{i-1}{2^n} \leq f(x) < \frac{i}{2^n}$ for some $i \in \{1, \ldots, n2^n\}$, and $f_n(x) = \frac{i-1}{2^n}$. Hence

$$0 \leq f(x) - f_n(x) < \frac{1}{2^n} \quad \text{for all } n \geq n_0,$$

and so $f_n(x) \to f(x)$ as $n \to \infty$.

Since x was arbitrary, $f_n \to f$ pointwise. //

Q5: Extra (B)

If $f, g \in L^1(\Omega, \mathcal{F}, \mu)$, show

$$\min(f, g) \in L^1(\Omega, \mathcal{F}, \mu)$$

and

$$\min\left(\int f\, d\mu, \int g\, d\mu\right) \geq \int \min(f, g)\, d\mu.$$

Answer

We utilize the following results:

- Definition 2.1.2(a) (measurable function): $f : \Omega \to \mathbb{R}$ is $\mathcal{F}$-measurable if $\{\omega : f(\omega) \leq a\} \in \mathcal{F}$ for each $a \in \mathbb{R}$.
- Definition 2.3.5 (integrability): a measurable f is in $L^1(\Omega, \mathcal{F}, \mu)$ iff $\int |f|\, d\mu < \infty$.
- Corollary 2.3.3: Proposition 2.3.1 holds for nonnegative measurable functions; in particular, $\int (\alpha f + \beta g)\, d\mu = \alpha \int f\, d\mu + \beta \int g\, d\mu$ for $\alpha, \beta \geq 0$, and $f \geq g$ a.e. (μ) implies $\int f\, d\mu \geq \int g\, d\mu$.
- Proposition 2.3.8(ii): for $f, g \in L^1(\Omega, \mathcal{F}, \mu)$, $f \geq g$ a.e. (μ) implies $\int f\, d\mu \geq \int g\, d\mu$.

Proof.

$\min(f, g) \in L^1(\Omega, \mathcal{F}, \mu)$**:** For each $a \in \mathbb{R}$, $\min(f, g)(\omega) \leq a$ iff $f(\omega) \leq a$ or $g(\omega) \leq a$, so

$$\{\omega : \min(f, g)(\omega) \leq a\} = \{\omega : f(\omega) \leq a\} \cup \{\omega : g(\omega) \leq a\} \in \mathcal{F},$$

and $\min(f, g)$ is $\mathcal{F}$-measurable (Definition 2.1.2(a)). At each $\omega \in \Omega$, $\min(f, g)(\omega)$ is either $f(\omega)$ or $g(\omega)$, so $|\min(f, g)| \leq |f| + |g|$. Then by Corollary 2.3.3,

$$\int |\min(f, g)|\, d\mu \leq \int \big(|f| + |g|\big)\, d\mu = \int |f|\, d\mu + \int |g|\, d\mu < \infty,$$

and so $\min(f, g) \in L^1(\Omega, \mathcal{F}, \mu)$ (Definition 2.3.5).

The inequality: Note that $\min(f, g) \leq f$ and $\min(f, g) \leq g$ on Ω. Since $\min(f, g), f, g \in L^1(\Omega, \mathcal{F}, \mu)$, Proposition 2.3.8(ii) gives

$$\int \min(f, g)\, d\mu \leq \int f\, d\mu \quad \text{and} \quad \int \min(f, g)\, d\mu \leq \int g\, d\mu,$$

and so $\int \min(f, g)\, d\mu \leq \min\big(\int f\, d\mu, \int g\, d\mu\big)$. //